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Experiment 1
Design of Star Schema and Snowflake Schema

Aim

To design Star Schema and Snowflake Schema for a given system using a suitable fact table and dimension tables.

Objective

- To understand the concept of Data Warehouse Schema.
- To design Star Schemas for two different subject areas of a college database.
- To design a Snowflake Schema by normalizing one of the star schemas.

Problem Statement

Consider a College Database System that maintains records related to student academic results and library book issues. The college wants to build a small data warehouse to analyze:

- How students are performing in different subjects and exams (Result Analysis).
- How books are being issued to students from the library (Library Management).

Since both areas need to be analyzed by Date, Student and other descriptive details, two separate Star Schemas are designed for these two subject areas. In addition, the Result Analysis star schema is further normalized into a Snowflake Schema by splitting the Student and Subject dimensions into smaller related tables (Department and Course).

Theory

1. Star Schema

A Star Schema centers on a fact table containing measurable data. The fact table connects directly to dimension tables that supply descriptive details about each record. These dimensions are usually denormalized, which keeps the structure simple and gives the diagram its familiar star-like shape.

2. Snowflake Schema

A Snowflake Schema extends the star design by normalizing selected dimension tables into related sub-tables. This separates repeated descriptive details, reduces redundancy, and creates more branches from the main dimensions. For example, student or subject details may be split into department or course tables, producing a snowflake-like structure.

Star Schema 1 — Student Result Analysis System

This schema is used to analyze the marks obtained by students in various subjects and exams over time.

Fact and Dimension Tables

Fact Table: Result_Fact

Column	Description
Result_ID	Primary Key
Student_ID	Foreign Key
Subject_ID	Foreign Key
Exam_ID	Foreign Key
Date_ID	Foreign Key
Marks_Obtained	Marks scored by the student
Max_Marks	Maximum marks for the exam

Student_Dimension

Column	Description
Student_ID	Primary Key
Student_Name	Name of the student
Class	Class of the student
Section	Section of the student

Subject_Dimension

Column	Description
Subject_ID	Primary Key
Subject_Name	Name of the subject
Subject_Type	Theory / Practical

Exam_Dimension

Column	Description
Exam_ID	Primary Key
Exam_Name	Name of the exam
Exam_Type	Internal / External
Semester	Semester in which exam is held

Date_Dimension

Column	Description
Date_ID	Primary Key
Date	Date of the exam
Month	Month
Quarter	Quarter
Year	Year

Diagram

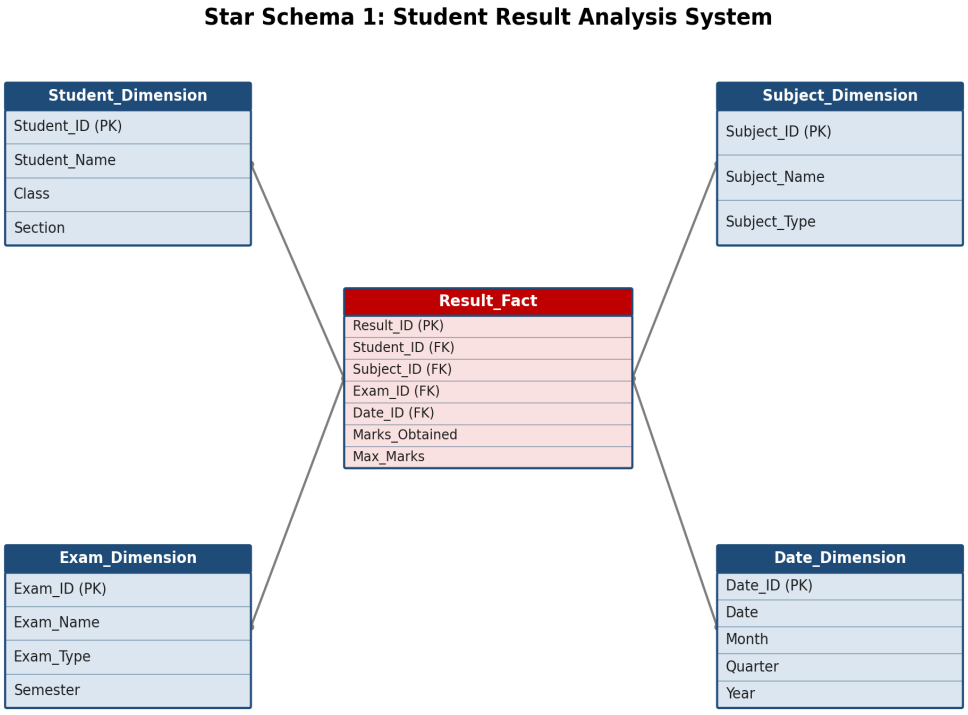


Fig 1.1: Star Schema for Student Result Analysis System

Star Schema 2 — Library Book Issue System

This schema is used to analyze how books are issued to students, by whom, and when.

Fact and Dimension Tables

Fact Table: Issue_Fact

Column	Description
Issue_ID	Primary Key
Book_ID	Foreign Key
Student_ID	Foreign Key
Staff_ID	Foreign Key
Date_ID	Foreign Key
Days_Issued	Number of days the book is issued for
Fine_Amount	Fine charged, if any

Book_Dimension

Column	Description
Book_ID	Primary Key
Book_Title	Title of the book
Author	Author of the book
Category	Category / Genre of the book

Student_Dimension

Column	Description
Student_ID	Primary Key
Student_Name	Name of the student
Class	Class of the student
Section	Section of the student

Staff_Dimension

Column	Description
Staff_ID	Primary Key
Staff_Name	Name of the library staff
Designation	Designation of staff

Date_Dimension

Column	Description
Date_ID	Primary Key
Date	Date of issue
Month	Month
Quarter	Quarter
Year	Year

Diagram

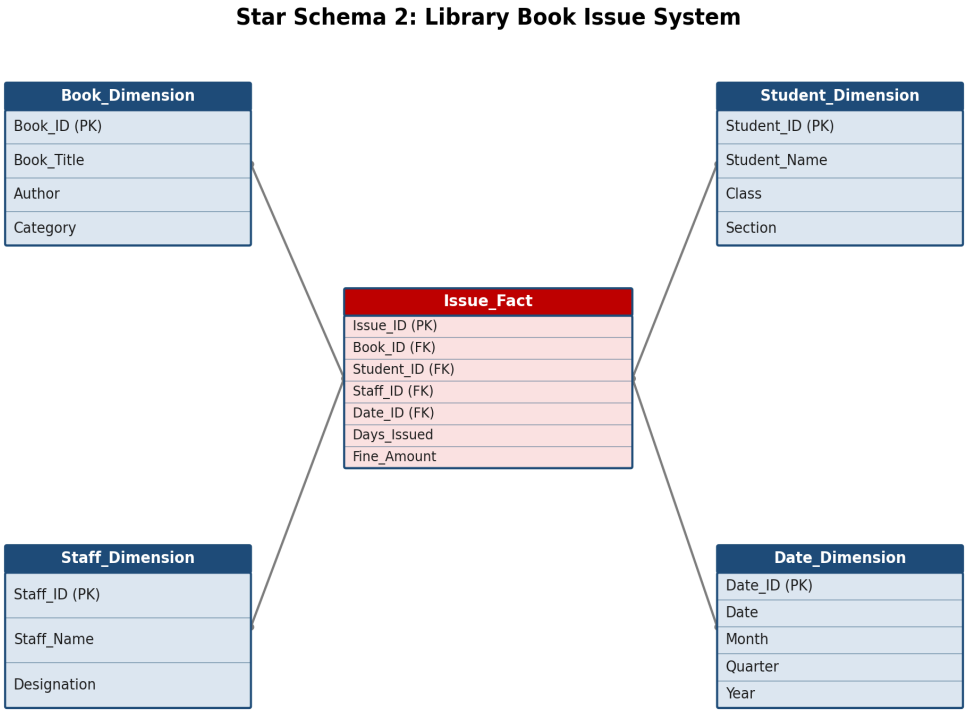


Fig 1.2: Star Schema for Library Book Issue System

Snowflake Schema — Student Result Analysis System (Normalized)

The Student Result Analysis star schema (Star Schema 1) is now normalized to form a Snowflake Schema. The Student_Dimension is split so that Department details move into a separate Department table, and the Subject_Dimension is split so that Course details move into a separate Course table. This removes repeated data and reduces redundancy.

Fact and Dimension Tables (Normalized)

Fact Table: Result_Fact

Column	Description
Result_ID	Primary Key
Student_ID	Foreign Key
Subject_ID	Foreign Key
Exam_ID	Foreign Key
Date_ID	Foreign Key
Marks_Obtained	Marks scored by the student
Max_Marks	Maximum marks for the exam

Student_Dimension

Column	Description
Student_ID	Primary Key
Student_Name	Name of the student
Dept_ID	Foreign Key to Department

Department (Sub-Dimension)

Column	Description
Dept_ID	Primary Key
Dept_Name	Name of the department
HOD_Name	Head of department

Subject_Dimension

Column	Description
Subject_ID	Primary Key
Subject_Name	Name of the subject
Course_ID	Foreign Key to Course

Course (Sub-Dimension)

Column	Description
Course_ID	Primary Key
Course_Name	Name of the course
Duration	Duration of the course

Exam_Dimension

Column	Description
Exam_ID	Primary Key
Exam_Name	Name of the exam
Exam_Type	Internal / External
Semester	Semester in which exam is held

Date_Dimension

Column	Description
Date_ID	Primary Key
Date	Date of the exam
Month	Month
Quarter	Quarter
Year	Year

Diagram

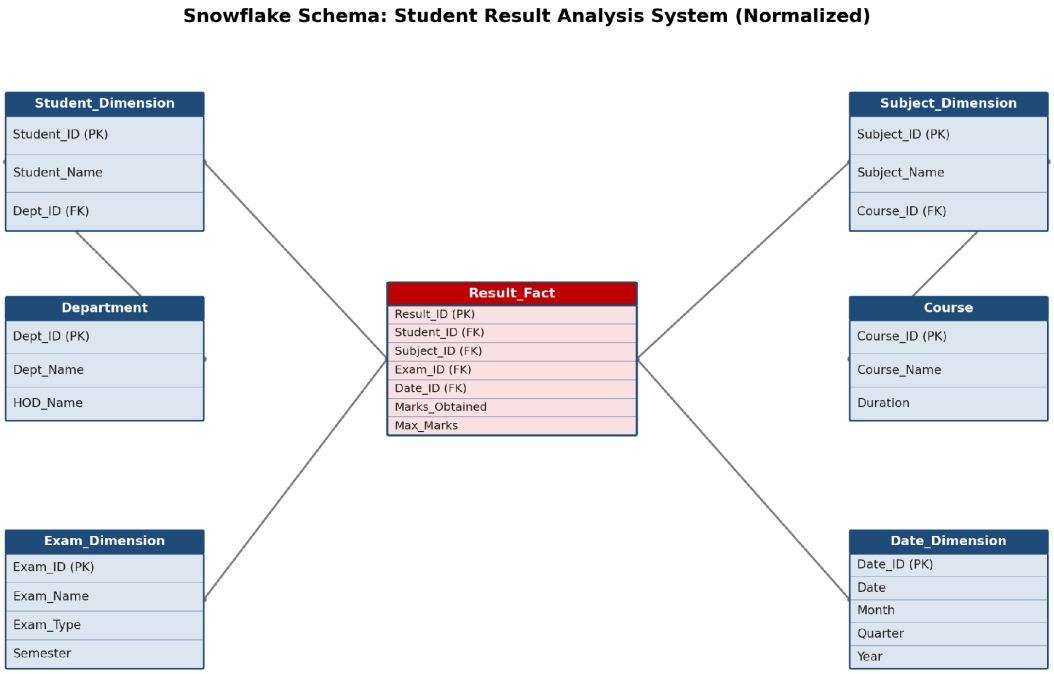


Fig 1.3: Snowflake Schema for Student Result Analysis System

Main Difference Between Star and Snowflake Schema

Star Schema	Snowflake Schema
Dimensions are generally denormalized	Dimensions are normalized
Fewer tables	More tables
Simple structure	More complex structure
Faster querying in many analytical cases	May require more joins
Requires more storage	Can reduce data redundancy

Result

Thus, two Star Schemas (Student Result Analysis and Library Book Issue) and one Snowflake Schema (normalized Student Result Analysis) were successfully designed by identifying suitable fact tables and dimension tables for the college database system.